

Assignment - 2

(1) line segment $(-5, 2)$ to $(25, 40)$

Parametric Equation,

$$\begin{aligned}P(t) &= P_0 + t \cdot (P_1 - P_0) \\&= (x_0, y_0) + t(x_1 - x_0, y_1 - y_0) \\&= (-5, 2) + t(25 - (-5), 40 - 2) \\&= (-5, 2) + t(30, 38) \\&= (-5, 2) + (30t, 38t) \\&= (-5 + 30t, 2 + 38t)\end{aligned}$$

Now, $t = 2/3$

$$\begin{aligned}P(2/3) &= (-5 + 30 \times 2/3, 2 + 38 \times 2/3) \\&= (15, \frac{82}{3})\end{aligned}$$

Again, For $t=5$, we need to verify if the point lies inside the line segment.

$$\begin{aligned}P(0) &= P_0 + 0 \cdot (P_1 - P_0) \\&= P_0 \\&= (x_0, y_0) \\&= (-5, 2)\end{aligned}$$

$$\begin{aligned}P(1) &= P_0 + 1 \cdot (P_1 - P_0) \\&= P_0 + P_1 - P_0 \\&= P_1 \\&= (x_1, y_1) \\&= (25, 40)\end{aligned}$$

→ can skip this part in your answer.

$$P(5) = (-5 + 30 \times 5, 2 + 38 \times 5) \\ = (145, 192)$$

$(145, 192)$ point doesn't lie inside the line segment $(-5, 2)$ to $(25, 40)$.

→ this part is enough for the last question.

The main roles -

- (b) 1. Helps to determine intersection points.
2. Then, the intersection points are grouped into two groups → PE and PL.

This is an important part of the algorithm.

[You can elaborate the points based on the algorithm.]

(c)

$$x_{\min} = -15$$

$$y_{\min} = 5$$

$$x_{\max} = 60$$

$$y_{\max} = 120$$

line segment $(20, 30)$ to $(90, 80)$
 $D = (90 - 20, 80 - 30) = (70, 50)$

Initially, $t_E = 0$, $t_L = 1$

Boundary	N_i	$N_i \cdot D$	t	PE/PL	t_E	t_L
Left	$(-1, 0)$	-70	-0.5	PE	0	1
Right	$(1, 0)$	70	0.57	PL	0	0.57
Bottom	$(0, -1)$	-50	-0.5	PE	0	0.57
Top	$(0, 1)$	50	1.8	PL	0	0.57

$P(0)$ and $P(0.57)$ are the true clip intersection.

$$\begin{aligned} P(0.57) &= (20, 30) + 0.57(70, 50) \\ &= (20, 30) + (39.9, 28.5) \\ &= (59.9, 58.5) \end{aligned}$$

$(20, 30)$ and $(59.9, 58.5)$ are the endpoints of the clipped line.

(2) (a) Already discussed in the class.

$$(b) \quad x_{\min} = -10, \quad x_{\max} = 20 \\ y_{\min} = -20, \quad y_{\max} = 30$$

line segment $(-10, -50)$ to $(15, 25)$

$$OC1 = \begin{array}{c} ABRL \\ 0100 \end{array}$$

$$OC2 = \begin{array}{c} 0000 \end{array}$$

$$OC1 \text{ AND } OC2 = 20000$$

The line is partially inside.

$$OC1 \neq 0000$$

OC1 has bottom bit.

Applying bottom intersection;

$$y_1 = y_{\min} = -20$$

$$x_1 = x_1 + \frac{1}{m} (y_{\min} - y_1) = -10 + \frac{1}{3} (-20 + 50)$$

$$P' = (0, -20) \quad z = 0$$

$$OC1 = 0000 \quad [\text{Recalculated}]$$

$$OC1 = z \quad OC2 = 0000$$

The clipped segment is $(0, -20)$ to $(15, 25)$